



Student's name: _____

Al-Kufa University
Engineering College



Electronic And Communication Department

3rd stage

LABORATORY MANUAL

Sequential Logic Circuit Lab

"Laboratory Experiments of Digital Logic Circuits"

Experiment 3

Shift Registers

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2023-2024

1. Aims:

- This experiment aims to provide a comprehensive understanding of the behavior of different kind of shift registers including their structure and working principle.

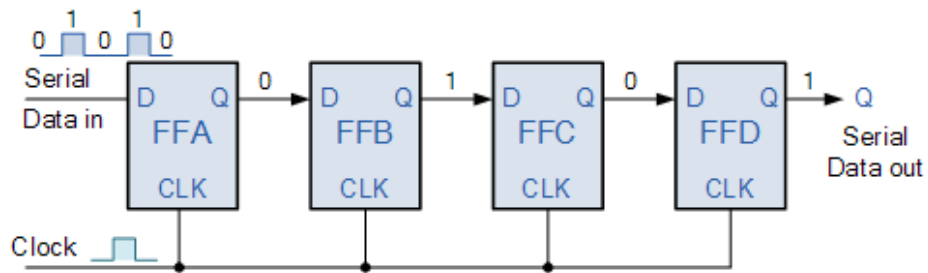
2. Theory :

The Shift Register is a type of sequential logic circuit that can be used for the storage or the transfer of binary data. Shift Registers are used for data storage or for the movement of data and are therefore commonly used inside calculators or computers to store data such as two binary numbers before they are added together, or to convert the data from either a serial to parallel or parallel to serial format. Shift registers hold the data in their memory which is moved or “shifted” to their required positions on each clock pulse. Each clock pulse shifts the contents of the register one bit position to either the left or the right.

Shift register IC’s are generally provided with a clear or reset connection so that they can be “SET” or “RESET” as required. Generally, shift registers operate in one of four different modes with the basic movement of data through a shift register being:

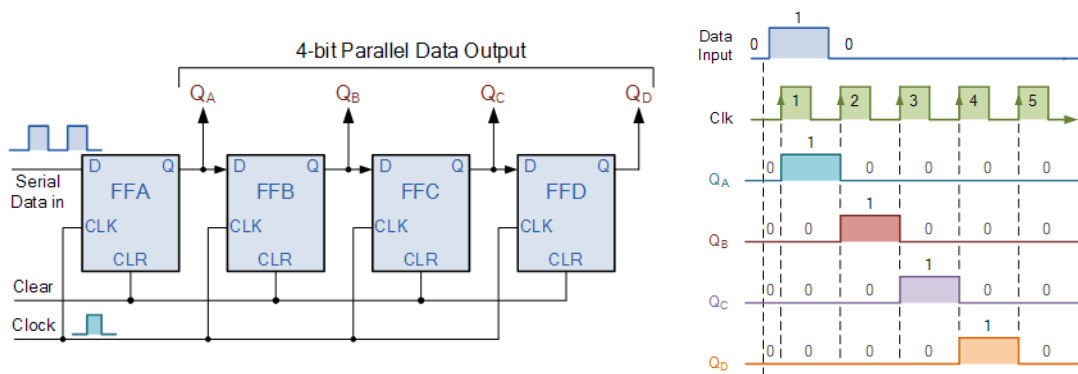
- Serial-in to Parallel-out (SIPO) – the register is loaded with serial data, one bit at a time, with the stored data being available at the output in parallel form.
- Serial-in to Serial-out (SISO) – the data is shifted serially “IN” and “OUT” of the register, one bit at a time in either a left or right direction under clock control.
- Parallel-in to Serial-out (PISO) – the parallel data is loaded into the register simultaneously and is shifted out of the register serially one bit at a time under clock control.
- Parallel-in to Parallel-out (PIPO) – the parallel data is loaded simultaneously into the register, and transferred together to their respective outputs by the same clock pulse.

2.1 (SISO):



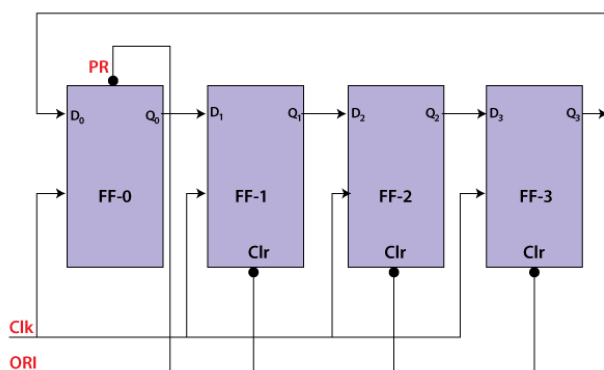
2.2 (SIPO):

Commonly available SIPO IC's include the standard 8-bit 74LS164 or the 74LS594.



2.2.1 Ring Counter:

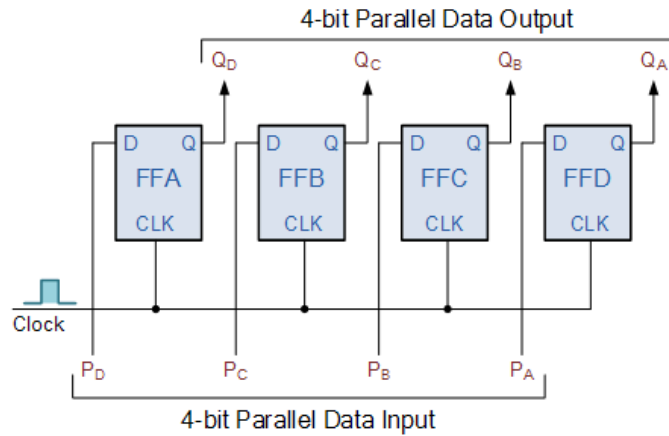
A ring counter is a special type of application of the Serial IN parallel OUT Shift register. The only difference between the shift register and the ring counter is that the last flip flop outcome is taken as the output in the shift register.



ORI	Clk	Q ₀	Q ₁	Q ₂	Q ₃
low	X	1	0	0	0
high	low	0	1	0	0
high	low	0	0	1	0
high	low	0	0	0	1
high	low	1	0	0	0

2.3 (PIPO):

This type of shift register also acts as a temporary storage device. one clock pulse loads and unloads the register.

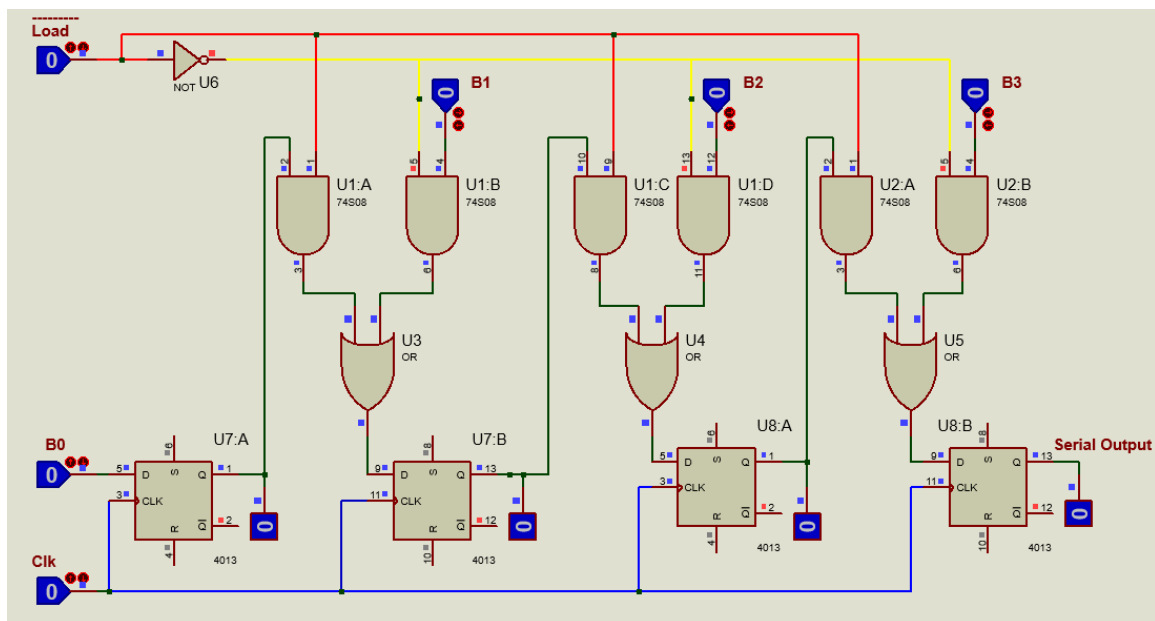


2.4 (PISO):

Commonly available IC's include the 74HC166 8-bit Parallel-in/Serial-out Shift Registers.

There are two working modes in this type of register:

- Load mode: to have the parallel input by giving logic 0 to the input called (Load)
- Shift mode: to have serial output by making the (Load) input high.

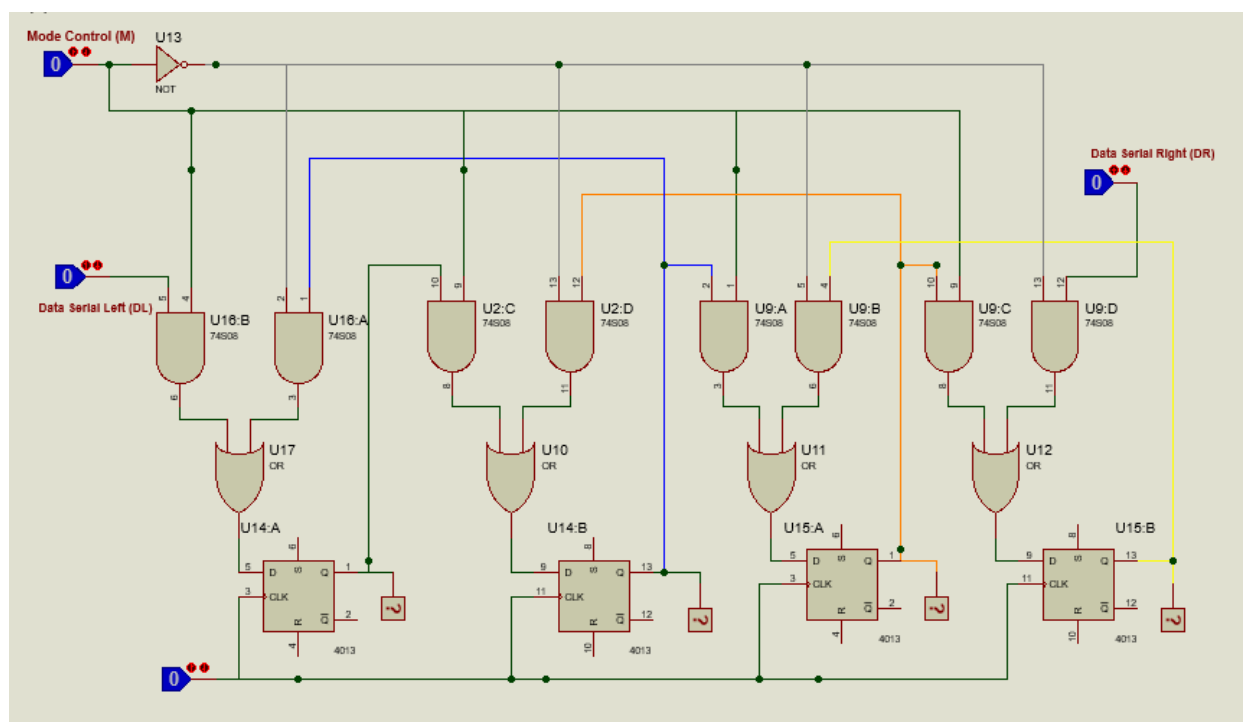


2.5 Bidirectional Shift Register:

A bidirectional shift register is a versatile circuit used for shifting data either to the right or left based on an input signal.

The bidirectional shift register operates based on clock pulses and control signals. It has two modes of operation: shift right and shift left. Let's explore these modes in detail:

1. **Shift Right:** In the shift right mode, the data is shifted from the leftmost bit to the rightmost bit. Each clock pulse triggers the transfer of data to the adjacent flip-flop on the right. The rightmost bit receives the input data, while the leftmost bit is discarded.
2. **Shift Left:** In the shift left mode, the data is shifted from the rightmost bit to the leftmost bit. Each clock pulse triggers the transfer of data to the adjacent flip-flop on the left. The leftmost bit receives the input data, while the rightmost bit is discarded.



3. Procedures:

3.1 Procedure 1:

1. Use Lab board to connect PISO shift register shown in section 2.4
2. Use data input as 0110 then fill the table below

Table 1

Load	Clk	Output			
		q_0	q_1	q_2	q_3
0	0				
0	1				
1	0				
1	1				
1	1				
1	1				
0	1				
1	1				

3. Comment on the results in Table1

3.2 Procedure 2:

1. Use Lab board to connect Bidirectional shift register shown in section 2.5
2. Fill the table below.

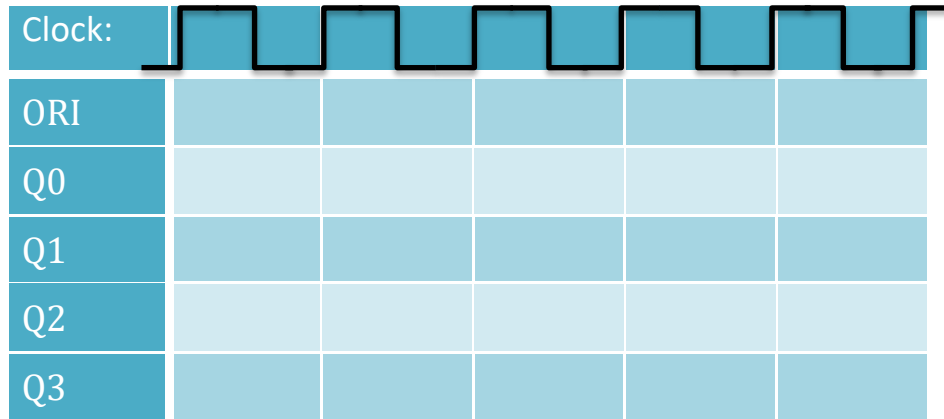
Table 2

M	DL	clk	DR	Output			
				q_0	q_1	q_2	q_3
1	1	0	0				
1	1	1	0				
1	0	1	0				
1	0	1	0				
1	0	1	0				
0	0	1	0				
0	0	1	0				
0	0	0	1				

3. Comment on the results in Table2

3.3 Procedure 3:

1. Use Lab board to connect the ring counter which is shown in section 2.2.1
2. Draw the output results as a waveforms on the figure below.



4. Discussions

1. Explain the working principles of universal shift register.
2. What are the applications of universal shift register and the bidirectional shift register?